E-Bank

Our proposed system is designed to help individuals and legal entities seamlessly manage bank accounts, cards, loans and financial operations through unified, fast, and secure transactions.

Problem

Banks operate using many different systems. They need to manage customers, issue cards, manage accounts, transfer money, understand their customers, and detect and prevent fraud. These systems are often developed or purchased at very different times and by different teams. Because they were not designed to work together from the beginning, integrating them and making them work together can be slow, complex, and messy. It can also make it harder to maintain and update the systems and introduce changes without breaking existing functionality.

Main Goal

Our system aims to unify these disjointed core banking, customer management, and compliance systems into a single, cohesive platform. By doing so, it simplifies financial management for customers, improves security, and eliminates delays in transaction processing and onboarding. In addition, one seamless banking system makes it easier for developers to maintain and modify functionality.

Core systems

- CBS (**Core Banking System)** - The backend system that processes daily financial transactions, updates account balances, manages deposits and loans, and maintains the central database for the bank.
- Payment account - Enables for the account to hold funds, receive deposits, and initiate local and international electronic fund transfers
- Accounting system - The financial tracking system that maintains general ledgers, manages double-entry bookkeeping, tracks balance sheets, and generates audit-ready financial statements.
- CRM (**Customer Relationship Management**) - Software used to manage customer data, track interactions, handle support tickets, and analyze customer behavior for marketing and sales.
- AML (**Anti-Money Laundering**) - Frameworks, regulations, and software systems designed to monitor transactions, detect suspicious activity, and prevent financial crimes.
- KYC (**Know Your Customer**) - The mandatory identity verification process required when onboarding new customers (e.g., verifying photo ID, proof of address, and risk profile) to prevent fraud and identity theft.
- Card management system - A comprehensive platform that handles the entire lifecycle of payment cards (debit, credit, prepaid), including card issuance, activation, PIN generation, transaction authorization, and blocking.
- Notification system - Sends real-time alerts via SMS, push notifications, and email for transactions, security updates, and account activities.

Team’s github: GitHub